

Probability – II: Assignment 1

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Exercise 1.

Problem

Let (X, Y) be jointly distributed with density

$$f(x, y) = \begin{cases} x + y & \text{if } x, y \in [0, 1] \\ 0 & \text{otherwise.} \end{cases}$$

Find the density of $(X^2 + Y, Y)$. Do not forget to mention the support.

Solution

Consider the transformation $(X, Y) \mapsto (X^2 + Y, Y)$. Let

$$U = X^2 + Y, \quad V = Y.$$

For any realization (u, v) we can solve

$$x = \sqrt{u - v}, \quad y = v.$$

Since x, y are uniquely recovered from (u, v) , the map is bijective, so we may

use the Jacobian (change-of-variables) formula. Thus, we have the following

$$J = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \begin{pmatrix} \frac{1}{2\sqrt{u-v}} & -\frac{1}{2\sqrt{u-v}} \\ 0 & 1 \end{pmatrix} \implies |\det J| = \frac{1}{2\sqrt{u-v}}.$$

Support. Since $x, y \in [0, 1]$:

$$v = y \in [0, 1], \quad x = \sqrt{u-v} \in [0, 1] \implies 0 \leq u-v \leq 1 \implies u \in [v, v+1].$$

Density.

$$f_{U,V}(u, v) = f(x(u, v), y(u, v)) \cdot |\det J| = (\sqrt{u-v} + v) \cdot \frac{1}{2\sqrt{u-v}} = \frac{1}{2} \left(1 + \frac{v}{\sqrt{u-v}} \right).$$

Hence

$$f_{U,V}(u, v) = \begin{cases} \frac{1}{2} \left(1 + \frac{v}{\sqrt{u-v}} \right) & \text{if } v \in [0, 1] \text{ and } u \in [v, v+1], \\ 0 & \text{elsewhere.} \end{cases}$$

Exercise 2.

Problem

If $X_1, \dots, X_{10}$ are iid $N(\mu, \sigma^2)$ where $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ are both unknown. Find $a > 0$ such that

$$\mathbb{P} \left(-a \leq \frac{\hat{\mu} - \mu}{\hat{\sigma}} \leq a \right) = 0.95, \quad \text{where } \hat{\mu} = \bar{X} \text{ and } \hat{\sigma}^2 = \frac{1}{9} \sum_{i=1}^{10} (X_i - \bar{X})^2.$$

Your answer should be in terms of $F_k(\cdot)$, the distribution function of the $t_{(k)}$ distribution.

Solution

Since $X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$ for $i = 1, \dots, 10$, and $\hat{\mu} = \bar{X}$, $\hat{\sigma}^2 = \frac{1}{9} \sum_{i=1}^{10} (X_i - \bar{X})^2$, we know

$$\hat{\mu} \sim N\left(\mu, \frac{\sigma^2}{10}\right), \quad \sum_{i=1}^{10} (X_i - \bar{X})^2 \sim \sigma^2 \chi_9^2 \implies \hat{\sigma}^2 \sim \frac{\sigma^2}{9} \chi_9^2 \implies \frac{9\hat{\sigma}^2}{\sigma^2} \sim \chi_9^2.$$

Also, since $\hat{\mu} \sim N\left(\mu, \frac{\sigma^2}{10}\right)$,

$$\sqrt{10} \frac{\hat{\mu} - \mu}{\sigma} \sim N(0, 1).$$

A t -distribution is the ratio of a standard normal and the square root of an independent chi-square divided by its degrees of freedom. Since $\hat{\mu}$ and $\hat{\sigma}^2$ are independent (and hence so are any functions of them), consider

$$Z = \frac{\sqrt{10} \frac{\hat{\mu} - \mu}{\sigma}}{\sqrt{\frac{9\hat{\sigma}^2}{\sigma^2} \cdot \frac{1}{9}}} = \sqrt{10} \frac{\hat{\mu} - \mu}{\hat{\sigma}} \sim t_9.$$

We are given

$$\mathbb{P}\left(-a \leq \frac{\hat{\mu} - \mu}{\hat{\sigma}} \leq a\right) = 0.95 \implies \mathbb{P}\left(-a\sqrt{10} \leq Z \leq a\sqrt{10}\right) = 0.95.$$

Thus

$$F_9(a\sqrt{10}) - F_9(-a\sqrt{10}) = 0.95.$$

Using $F_9(-x) = 1 - F_9(x)$,

$$F_9(a\sqrt{10}) - [1 - F_9(a\sqrt{10})] = 0.95 \implies F_9(a\sqrt{10}) = \frac{1.95}{2} = 0.975.$$

Therefore

$$a = \frac{F_9^{-1}(0.975)}{\sqrt{10}}$$

where F_9^{-1} is the inverse distribution function (quantile function) of t_9 . Numerically, $F_9^{-1}(0.975) \approx 2.262$, so

$$a \approx \frac{2.262}{\sqrt{10}} \approx \frac{2.262}{3.1622} \approx 0.7154.$$

Exercise 3.

Problem

A branching process has progeny distribution

$$\mathbb{P}(X = k) = \begin{cases} \frac{1}{5} & \text{if } k = 0, 1, 2, 3, 4 \\ 0 & \text{otherwise.} \end{cases}$$

Show that the extinction probability is < 1 .

Solution

Mean argument. The expected number of offspring is

$$\mathbb{E}(X) = \sum_{k=0}^4 k \cdot \mathbb{P}(X = k) = \frac{1}{5}(0 + 1 + 2 + 3 + 4) = 2.$$

Since the expected number of offspring exceeds 1, this guarantees that the extinction probability is < 1 .

Alternative (generating function) argument. Recall that the extinction probability θ is the smallest solution in $[0, 1]$ of $g(t) = t$, where g is the progeny generating function, and $g(t) = t$ has at most two solutions in $[0, 1]$, one of which is always $t = 1$. Here

$$g(t) = \sum_{k=0}^4 t^k \mathbb{P}(X = k) = \frac{1}{5} (1 + t + t^2 + t^3 + t^4).$$

Setting $g(t) = t$ and defining

$$h(t) = g(t) - t = \frac{1}{5} (1 - 4t + t^2 + t^3 + t^4) = 0 \iff 1 - 4t + t^2 + t^3 + t^4 = 0.$$

Clearly $h(1) = 0$ and $h(0) = 1$. Hence there exists $\theta \in (0, 1)$ with $h(\theta) = 0$. Since $\theta < 1$ and θ is the smallest root of $g(t) = t$ in $[0, 1]$, this θ is exactly the extinction probability.

$\implies$ extinction probability < 1 . ■

Exercise 4.

Problem

If X has $N_3(\boldsymbol{\mu}, \Sigma)$ distribution, where

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix},$$

then prove or disprove the statement:

$X_1 + X_2 + X_3$ and $X_1 - 2X_2 + X_3$ are independent.

Solution

Since $X \sim N_3(\boldsymbol{\mu}, \Sigma)$, we have $X_i \sim N(\mu_i, \sigma_i^2)$ for $i = 1, 2, 3$, with

$$\mathbb{E}(X_1) = 1, \mathbb{E}(X_2) = 2, \mathbb{E}(X_3) = 3, \quad \text{Var}(X_1) = 1, \text{Var}(X_2) = 2, \text{Var}(X_3) = 3,$$

and $\text{Cov}(X_i, X_j) = 0$ for $i \neq j$ (since Σ is diagonal).

Let

$$U = X_1 + X_2 + X_3, \quad V = X_1 - 2X_2 + X_3.$$

Since U and V are linear combinations of jointly normal random variables, (U, V) is itself jointly normal; hence U and V are independent *iff* $\text{Cov}(U, V) = 0$.

$$\text{Cov}(U, V) = \text{Cov}(X_1 + X_2 + X_3, X_1 - 2X_2 + X_3).$$

Expanding and using $\text{Cov}(X_i, X_j) = 0$ for $i \neq j$:

$$\text{Cov}(U, V) = \text{Var}(X_1) - 2\text{Var}(X_2) + \text{Var}(X_3) = 1 - 2(2) + 3 = 0.$$

Since $\text{Cov}(U, V) = 0$ and (U, V) is jointly normal, U and V **are independent.** ■

Exercise 5.

Problem

Let X be any nonnegative random variable with finite third moment (i.e., $\mathbb{E}(X^3)$ is a finite number). Then show that

$$\forall \varepsilon > 0, \quad \mathbb{P}(X \geq \varepsilon) \leq \frac{\mathbb{E}(X^3)}{\varepsilon^3}.$$

Solution

Markov's Inequality. For any nonnegative random variable Y and any $a > 0$,

$$\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}(Y)}{a}. \quad (1)$$

Since $g(x) = x^3$ is monotonically increasing for $x \geq 0$, and $X \geq 0$,

$$\mathbb{P}(X \geq \varepsilon) = \mathbb{P}(X^3 \geq \varepsilon^3).$$

Now apply (1) with $Y = X^3$ and $a = \varepsilon^3$:

$$\mathbb{P}(X^3 \geq \varepsilon^3) \leq \frac{\mathbb{E}(X^3)}{\varepsilon^3}.$$

Therefore

$$\mathbb{P}(X \geq \varepsilon) \leq \frac{\mathbb{E}(X^3)}{\varepsilon^3}. \quad \blacksquare$$